

Find missing dimensions

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: A triangle has area 54 cm^2 and base 12 cm . Find its perpendicular height. Copy or complete the first step.

2. A shape combines a 10 cm by 6 cm rectangle and a triangle with base 10 cm and height 4 cm . Find total area.

3. A rectangle is 14 cm by 9 cm . Find its area and perimeter.

4. Miro says: Miro uses the sloping side as the height and forgets to halve a triangle's area. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: A triangle has area 54 cm² and base 12 cm. Find its perpendicular height. Copy or complete the first step.
Answer 9 cm.
2. A shape combines a 10 cm by 6 cm rectangle and a triangle with base 10 cm and height 4 cm. Find total area.
Answer 80 cm².
3. A rectangle is 14 cm by 9 cm. Find its area and perimeter.
Answer Area 126 cm²; perimeter 46 cm.
4. Miro says: Miro uses the sloping side as the height and forgets to halve a triangle's area. Is this correct? Explain.
Answer Mark the perpendicular height, use base times height, and divide by two only for a triangle.

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Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. A triangle has area 54 cm^2 and base 12 cm . Find its perpendicular height.

6. Check this result using an inverse, estimate or known fact: A rectangle is 14 cm by 9 cm . Find its area and perimeter.

2. A shape combines a 10 cm by 6 cm rectangle and a triangle with base 10 cm and height 4 cm . Find total area.

7. Prove or explain your reasoning: Find the area of a parallelogram with base 12 cm and perpendicular height 7 cm .

3. A rectangle is 14 cm by 9 cm . Find its area and perimeter.

8. Identify and correct this misconception: Miro uses the sloping side as the height and forgets to halve a triangle's area.

4. Solve this independently and show every stage: A triangle has area 54 cm^2 and base 12 cm . Find its perpendicular height.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: A shape combines a 10 cm by 6 cm rectangle and a triangle with base 10 cm and height 4 cm . Find total area.

10. Write and solve a similar question to: A rectangle is 14 cm by 9 cm . Find its area and perimeter.

Teacher answers and checking prompts

1. A triangle has area 54 cm² and base 12 cm. Find its perpendicular height.
Answer 9 cm.
2. A shape combines a 10 cm by 6 cm rectangle and a triangle with base 10 cm and height 4 cm. Find total area.
Answer 80 cm².
3. A rectangle is 14 cm by 9 cm. Find its area and perimeter.
Answer Area 126 cm²; perimeter 46 cm.
4. Solve this independently and show every stage: A triangle has area 54 cm² and base 12 cm. Find its perpendicular height.
Answer 9 cm.
5. Use a second method or representation for: A shape combines a 10 cm by 6 cm rectangle and a triangle with base 10 cm and height 4 cm. Find total area.
Answer Expected result: 80 cm². Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: A rectangle is 14 cm by 9 cm. Find its area and perimeter.
Answer Expected result: Area 126 cm²; perimeter 46 cm. A valid check must be shown.
7. Prove or explain your reasoning: Find the area of a parallelogram with base 12 cm and perpendicular height 7 cm.
Answer 84 cm². The explanation must show why the method works.
8. Identify and correct this misconception: Miro uses the sloping side as the height and forgets to halve a triangle's area.
Answer Mark the perpendicular height, use base times height, and divide by two only for a triangle.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: A rectangle is 14 cm by 9 cm. Find its area and perimeter.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Find missing dimensions

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: Find the area of a parallelogram with base 12 cm and perpendicular height 7 cm.

6. Create a new example that tests the same idea as: A rectangle is 14 cm by 9 cm. Find its area and perimeter.

2. Prove the answer to this question, rather than only calculating it: A triangle has area 54 cm² and base 12 cm. Find its perpendicular height.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: A shape combines a 10 cm by 6 cm rectangle and a triangle with base 10 cm and height 4 cm. Find total area.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: Area 126 cm²; perimeter 46 cm.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro uses the sloping side as the height and forgets to halve a triangle's area.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Find the area of a parallelogram with base 12 cm and perpendicular height 7 cm.
Answer 84 cm². The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: A triangle has area 54 cm² and base 12 cm. Find its perpendicular height.
Answer Expected result: 9 cm. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: A shape combines a 10 cm by 6 cm rectangle and a triangle with base 10 cm and height 4 cm. Find total area.
Answer Expected result: 80 cm². Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: Area 126 cm²; perimeter 46 cm.
Answer One valid reconstruction is: A rectangle is 14 cm by 9 cm. Find its area and perimeter. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro uses the sloping side as the height and forgets to halve a triangle's area.
Answer Mark the perpendicular height, use base times height, and divide by two only for a triangle.
6. Create a new example that tests the same idea as: A rectangle is 14 cm by 9 cm. Find its area and perimeter.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.